1.7 Safety Devices

1.7.1 Air Bag

An airbag is a type of vehicle safety device and is an occupant restraint system. The airbag module is designed to inflate extremely rapidly and then quickly deflate during a collision or impact with a surface or a rapid sudden deceleration. It consists of the airbag cushion, a flexible fabric bag, inflation module and impact sensor. The purpose of the airbag is to provide the occupants with a soft cushioning and restraint during a crash to prevent any impact or impact-caused injuries between the failing occupant and the interior of the vehicle like steering wheel, instrumental panel, structural body frame, headliner and windshield. Refer Figure 1.7.1.



Figure 1.7.1 Airbag

1.7.2 Anti-Lock Braking System (ABS)

An anti-lock braking system (ABS) is an automobile safety system that allows the wheel to maintain tyre contact with the road surface while sudden braking. It prevents the wheels from locking up and it will avoid skidding of vehicles. Refer Figure 1.7.2.

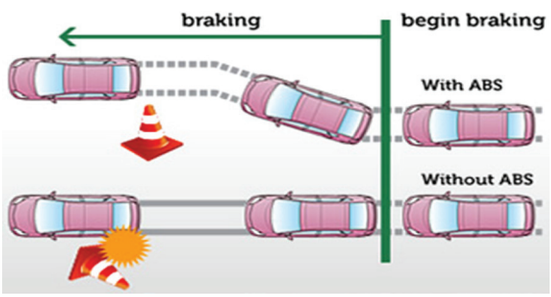


Figure 1.7.2 Anti-lock Braking System (ABS)

1.7.3 Automatic Door Lock

Automatic door locks will lock all the door simultaneously if the driver forgets to lock the doors. This will increase the safety of the car as well as things kept inside the car.

1.7.4 Steering Wheel Lock

Modern vehicles are fitted with a steering lock and it is an anti-theft device. It is fitted to the steering column usually below the steering wheel. The lock is combined with the ignition switch and engaged and disengaged either by a mechanical ignition key or electronically from the vehicles electronic control unit.

1.8 FIRST AID

Even a person following safety rules, there may be chances of an accident due to sudden failure of hand tools or machines. In such case, the medical assistance given to such person before taking to the hospital is called as First aid. First aid is provided to preserve life, prevent the condition from worsening or to promote recovery. First aid is generally performed by the layperson, with training in basic levels of first aid.

First Aid was started by S. Mark in the year 1823. A First Aid Kit should have the following contents. Shown in Figure 2(a).

1. Tincture Iodine
2. Tincture Benzene
3. Dettol
4. Burnol
5. Boric Powder
6. Meshed Cloth

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